8	**(a)**		$\frac{dy}{dx} = 3x^2$	**M1**	**3.1a**		
			Gradient at (1, 1) is 3	**M1**	**1.1**		
			Tangent is $(y-1) = 3(x-1)$	**A1**	**1.1**	($y = 3x - 2$; simplified form not essential)	**or** $\frac{AB}{AD} = 3$
			OE = 2 and OD = $\frac{2}{3}$	**M1**	**3.1a**	Finding either of these lengths soi by E (0, -2) or D ($\left(\frac{2}{3}, 0\right)$	**or** AD = $\frac{1}{3}$
				A1	**1.1**	Bothe OE and OD correct	**Or** Triangle ODE is similar to ABD [with scale factor 2]
			Area triangle ODE = $\frac{1}{2} \times \frac{2}{3} \times 2 = \frac{2}{3}$	**A1**	**1.1**		
				[6]			

8	**(b)**		**DR** Area $= \int_0^1 \left(x^3 - (3x-2)\right)\,\mathrm{d}x$	**M1** **M1**	**3.1a** **1.1**	Integrand. FT their tangent Limits (may be seen later)	
			$\left[\frac{x^4}{4} - \frac{3x^2}{2} + 2x\right]_0^1$	**M1**	**1.1**	Allow one error	
			$\frac{3}{4}$	**A1**	**2.2a**		
			Alternative method Area between curve and x-axis = $\int_0^1 x^3\,\mathrm{d}x = \left[\frac{x^4}{4}\right]_0^1 = \left[\frac{1}{4}\right]$	**M1**			
			Area triangle ABD $= \frac{1}{2}\times\frac{1}{3}\times 1 = \frac{1}{6}$	**M1**			
			Area = area between curve and x-axis + triangle ODE – triangle ABD	**M1**		Using *their* values	
			$\frac{3}{4}$	**A1**			
				[4]			